

845 Sherbrooke Street Ouest
Montréal, Québec, H3A 0G4
mohamed.debbagh@mail.mcgill.ca
5145692537
2024/12/10

Editorial Office
Plant Phenomics
Ref.: Ms. No. PlantPhenomics-D-25-00006

Dear Editors and Reviewers,

We have received your comments for our manuscript submitted under the title “Predictive Pattern Recognition Techniques Towards Spatiotemporal Representation of Plant Growth in Simulated and Controlled Environments: A Comprehensive Review”. We would first like to thank you for your time and consideration for reviewing the work. The comments were thorough and demonstrate your expertise within the field and are much appreciated. Please note based on the comments we have decided to retitile the manuscript to **“Predictive Modeling, Pattern Recognition, and Spatiotemporal Representations of Plant Growth in Simulated and Controlled Environments: A Comprehensive Review”**. We have made major revisions to the manuscript based on your comments. You can find each point for each reviewer addressed with corresponding revisions, or rebuttals made in the following pages attached to this cover letter.

We believe these revisions make the paper stronger and further aligns with the scope of the Plant Phenomics journal and will contribute to ongoing discussions in the field. If you have any further questions, comments or would like to have discussions for future works, please do not hesitate to reach out.

Sincerely,

Mohamed Debbagh,
On behalf of all authors

To Reviewer #1:

Comment 1: Defocus on Simulated and Controlled Environments - The scope of this review paper is limited to simulated and controlled environments. Compared to the common open field environments in agricultural production and research, controlled environments improve agricultural availability and productivity by reducing environmental variability and intensifying management practices. To establish the context for your review and analysis, you need to explain the differences between open field environment and controlled environment. Subsequently, you can introduce the techniques applied in controlled environments and analyze why those techniques are applicable and effective in controlled environments.

- Response to Comment 1 (Revision): This is a good point, we did not clarify the implications of controlled environments within the scope of this review. To address this, we have replaced the background subsection with 3 subsections, one of which is titled: 1.1 Comparing Controlled and Open Field Environments. This subsection adds context to the focus of the review within the introduction section and introduces the differences between open field and controlled environments and their implications towards modeling with added references.
- **Revisions made correspond to lines 39~65 in the manuscript.**

Comment 1-continued 1: Section 3.1 discussed regression models developed from open field experiments. Those models have fixed variables and configurations which is a shortcoming due to the inflexibility in dynamic open field conditions. However, the stability of controlled environments might contribute to better performance of those fixed-variable models. You need to discuss the potential interactions between controlled environments and the fixed-variable models.

- Response to Comment 1-continued 1 (Revision): To address this point, we have added subsection 3.1.2 How Controlled Environments Affect Fixed Variables in Regressions. In this subsection we discuss the potential interactions between controlled environments and the fixed-variable models.
- **Revisions made correspond to lines 275~293 in the manuscript.**

Comment 1-continued 2: I am confused about the simulated environment. The term, simulated environment or simulated agriculture, has dual interpretations. One is the controlled environment that simulates the optimal growth conditions for plants. The other is the cyber-agricultural system which digitalizes the plant status from physical world to virtual cyberspace and then optimizes management practices based on crop model output. Please clarify which definition applies in this context.

- Response to Comment 1-continued 2 (Revision): To address this point (and partially the next Comment 2), we have replaced the background subsection with 3 subsections, one

of which is titled: 1.2 Defining Simulated Environments and Plant Growth. In this subsection we provide clarifications between the terms Simulation, and plant growth within the context of controlled environments which in turn is a means to mimic real-world field conditions.

- Revisions made correspond to lines 66~91 in the manuscript.

Comment 2: Insufficient Definition of Plant Growth - There are so many indicators evaluating plant growth and development, some are quantitative, some are empirical, some are physiological, some are morphological, etc. Plant growth is a very broad scope for a single paper. Your review paper will be more comprehensive if you narrow the domain to which type of plant growth indicators you want to review, specify the major indicators or traits, and then analyze the measurement, modeling and prediction of them in-depth. Besides, I recommend you learn some numerical crop models, like APSIM, AquaCrop, DSSAT, WOFOST, Hybrid-Maize. Although they don't model the single trait directly, they provide a systematic framework to describe and calculate plant growth.

- Response to Comment 2 (Revision): Clarifications for defining plant growth are made in the added subsection: 1.2 Defining Simulated Environments and Plant Growth. Thank you for the suggestions on looking into crop models. Numerical Crop models typically combine various empirical models and crop specific mechanistic models of different processes within the plant growth cycle and at multiple levels. We believe this may be too large and beyond the scope of this review. For this review we wanted to focus namely on fundamental principles of spatiotemporal pattern recognition and empirical methods for predicting plant traits through time and different representations of these spatiotemporal prediction models. However we agree that the ultimate goal will be to incorporate these into larger crop model frameworks.
- Revisions made correspond to lines 66~91 in the manuscript.

Comment 3: Disorganized Framework and Categorization- Since your scope is too large, the framework and categorization are disorganized. The section 3 and 4, temporal modeling paradigms of quantitative traits and 2D/3D structured representations through time, have significant overlap. Please consider the general pipeline for phenotyping. There are three major steps in a pipeline:

a). manual ground truth collection, b). digital data acquisition, c). model training for quantitative trait prediction

We can collect 2D/3D data easily. Why do we need to train the models for phenotyping? Because those 2D/3D data are high dimensional that we can't evaluate plant growth directly. We still need

some low-dimensional indicators, like scalar in your paper but I'd like to call them quantitative traits, to assess plant growth. Please decouple the overlap to have a clear framework.

- Response to Comment 3 (Revision & Rebuttal):
- (Rebuttal 1) The modeling pipeline (phenotyping pipeline) is described and discussed in section 1.3 Background: Predictive Plant Growth Modeling and further aided by Figure 1. In response to needing low-dimensional traits: Not necessarily, only if you want to couple model outputs to a ground truth quantitative trait, then yes, quantitative traits must be collected alongside. This is no different than making any trait prediction model and not necessarily unique to spatiotemporal pattern recognition. However, if your simulation output is a structured output over time (or pattern identification), such as a sequence of anomalies, generated images or a 3D structures that evolve through time based on environmental parameters, then the “ground truth” can be a structured data like a real image (+ a prior state) labeled with a timestamp and the environmental parameters collected and not a quantitative traits. The models in section 4 perform pattern recognition on structure data through time rather than ground truth quantitative traits.
- (Rebuttal 2) Section 3 and 4 may have some overlaps since section 4 doesn't completely divorce these concepts. This is pointed out in line 469 ~ 470 where we mention “In this section, we take the foundation of pattern recognition techniques explored in Section 3 and apply them toward tasks that represent plant morphologies as structured outputs.”. The methodologies in this paper are organized as such: The paper introduces temporal models for quantitative traits in section 3, and then we introduce structured representations and models that apply temporal modeling paradigms to these structured outputs for visual representation and pattern recognition through time in section 4.
- (Revision) I have made some clarifications within the scope of this review to make this point clearer and added it as section 1.4 Scope of this Review
- **Revisions made correspond to lines 143~159 in the manuscript.**
- (Revision) At the end of each section (3.3 and 4.3) I have added a comparison subsection and table which compares the model's data requirements, uncertainty handling, interpretability and application, to further clarify differences between models.
- **Revisions made correspond to lines in the manuscript:**
 - **Section 3.3: lines 448~461**
 - **+ Table 3: Page 17**
 - **Section 4.3: lines 624~639**
 - **+ Table 4: Page 25**

Comment 4: Weak Reference Paper- Only one or two papers were cited in many subsections. The limited reference is not strong enough to support your analysis and opinion, especially in the section of Perspectives.

- Response to Comment 4 (Revisions): Apologies, while the assessment of perspectives was based on the consensus of literature, we did not properly add enough citations especially in core statements within our opinions. To address this we have gone over the perspectives section, modified statements that required literature backing and properly added references. In addition, where possible we provided literature examples of applications. Note on some methodologies: we have added a few more references to the analysis of frameworks for some methodologies. However please note that many of these works are considered SOTA and the field is still early in its development with limited availability of works. To make this point clearer we have added to the limitation section of the scope of this paper, describing limitations in available papers that discuss novel methods.
- Revisions to limitation statement are found in lines 198~208
- The following references were added (reference number, title, citation in manuscript):
 - [13] Plant phenomics and the need for physiological phenotyping across scales to narrow the genotype-to-phenotype knowledge gap. Cited in line 51
 - [14] Field-based phenomics for plant genetics research. Cited in line 51
 - [15] Field Phenomics: Will It Enable Crop Improvement?. Cited in line 54
 - [16] Field crop phenomics: enabling breeding for radiation use efficiency and biomass in cereal crops. Cited in line 54
 - [17] Field high-throughput phenotyping: the new crop breeding frontier. Cited in line 54
 - [18] A controlled environment agriculture with hydroponics: variants, parameters, methodologies and challenges for smart farming. Cited in line 61
 - [19] Predicting Genotype x Environment x Management (Gx Ex M) interactions for the design of crop improvement strategies: integrating breeder, agronomist, and farmer perspectives. Cited in line 69
 - [20] Leveraging biological insight and environmental variation to improve phenotypic prediction: Integrating crop growth models (CGM) with whole genome prediction (WGP). Cited in line 69
 - [21] Genotype x Environment x Management (GEM) reciprocity and crop productivity. Cited in line 69
 - [22] Plant Growth and Development - Basic Knowledge and Current Views. Cited in line 86
 - [23] Plant development. Cited in line 86
 - [24] Future Scenarios for Plant Phenotyping. Cited in line 89

- [75] Multi environments and genetic-environmental interaction (GxE) in plant breeding and its challenges: a review article. Cited in line 285
- [76] Spatial regression models for field trials: a comparative study and new ideas. Cited in line 290
- [77] Collinearity: a review of methods to deal with it and a simulation study evaluating their performance. Cited in line 293
- [78] What you see may not be what you get: a brief, nontechnical introduction to overfitting in regression-type models. Cited in line 303
- [79] The temporal overfitting problem with applications in wind power curve modeling. Cited in line 303
- [80] On overfitting and post-selection uncertainty assessments. Cited in line 305
- [81] Optimal inference after model selection. Cited in line 305
- [82] Universally sloppy parameter sensitivities in systems biology models. Cited in line 312
- [83] Time series analysis and its applications. Cited in line 318
- [86] Growth dynamics and heritability for plant high-throughput phenotyping studies using hierarchical functional data analysis. Cited in line 335
- [87] Forecast evaluation for data scientists: common pitfalls and best practices. Cited in line 346
- [98] The mythos of model interpretability: In machine learning, the concept of interpretability is both important and slippery. Cited in line 419
- [99] Interpretability and the nature of latent representations in deep neural networks. Cited in line 419
- [100] Toward transparent ai: A survey on interpreting the inner structures of deep neural networks. Cited in line 421
- [101] Principal component analysis: a review and recent developments. Cited in line 426
- [104] Deep Learning Meets Process-Based Models: A Hybrid Approach to Agricultural Challenges. Cited in lines 447 and 663
- [116] Mathematical models for cellular interactions in development I. Filaments with one-sided inputs. Cited in line 513
- [138] A time-dependent parameter estimation framework for crop modeling. Cited in line 644
- [139] Literature Review Crop Modeling and Introduction a Simple Crop Model. Cited in line 644
- [140] Challenges and opportunities in crop simulation modelling under seasonal and projected climate change scenarios for crop production in South Africa. Cited in line 641
- [142] How to make sense of 3D representations for plant phenotyping: a compendium of processing and analysis techniques. Cited in lines 651 and 729

- [143] Deep learning to capture leaf shape in plant images: Validation by geometric morphometrics. Cited in line 651
- [144] Explainable deep learning in plant phenotyping. Cited in line 654 and 665
- [145] Embracing limited and imperfect training datasets: opportunities and challenges in plant disease recognition using deep learning. Cited in line 669
- [146] Resources for image-based high-throughput phenotyping in crops and data sharing challenges. Cited in line 669
- [147] Label-efficient learning in agriculture: A comprehensive review. Cited in line 670
- [148] Analysing the phenotype development of soybean plants using low-cost 3D reconstruction. Cited in line 673
- [149] A Review of Imaging Techniques for Plant Phenotyping. Cited in line 680
- [150] Challenges for FAIR-compliant description and comparison of crop phenotype data with standardized controlled vocabularies. Cited in line 685
- [151] Making deep neural networks right for the right scientific reasons by interacting with their explanations. Cited in line 695
- [152] Anomaly Detection for Internet of Things Time Series Data Using Generative Adversarial Networks With Attention Mechanism in Smart Agriculture. Cited in line 710
- [153] Tranad: Deep transformer networks for anomaly detection in multivariate time series data. Cited in line 713
- [154] Anomaly transformer: Time series anomaly detection with association discrepancy. Cited in line 713
- [155] A single view leaf reconstruction method based on the fusion of ResNet and differentiable render in plant growth digital twin system. Cited in line 729
- [156] A 3D functional plant modelling framework for agricultural digital twins. Cited in line 735

To Reviewer #2:

Comment 1: Consider labeling the Introduction as Section 1, making "Rationale and Approach" Section 2 (with sub-sections 2.1, 2.2), and then numbering Sections 3 and 4 sequentially so the structure feels clear and consistent.

- Response to Comment 1 (Point of clarification/ Rebuttal): We believe that the paper is already organized as such with the section numbering. Please let us know if we misunderstood this point!

Comment 2: Adding a brief roadmap paragraph at the end of the Introduction (or at the start of Section 2) that previews the major sections—regression methods, latent state models, neural network approaches, functional-structural plant models, and computer-vision simulations—will help readers see the overall flow before diving into details.

- Response to Comment 2 (Revision): We have added section 1.4 which outlines the scope of the project, and provides a simple outline. We had originally put it in section 2 but it is better placed in section 1 to serve as a roadmap.
- **Revisions made correspond to lines 143~159 in the manuscript.**

Comment 3: After each major category of methods, you might include a short summary paragraph or a simple table comparing data requirements, how uncertainty is handled, interpretability, and typical applications. A side-by-side comparison will highlight trade-offs and guide readers in selecting the right approach.

- Response to Comment 3 (Revision): We have added sections 3.3 Comparisons of temporal Models and 4.3 Comparisons of Structured Output Representations Through Time as well as Table 3 and Table 4 to highlight comparisons side by side.
- **Revisions made correspond to lines in the manuscript:**
 - Section 3.3: lines 448~461
 - + Table 3: Page 17
 - Section 4.3: lines 624~639
 - + Table 4: Page 25

Comment 4: For the conditional generative models section, a few sentences explaining how latent sampling guides the image synthesis process will deepen readers' understanding.

- Response to Comment 4 (Revision): We have added a sentence explaining latent sampling in section 4.2.2.
- **Revisions made correspond to lines 584~589 in the manuscript.**

Comment 5: When introducing "digital twin," you could define it in one sentence (for example, "a virtual replica of a plant system updated in real time with sensor data") and give a short example—such as its use in greenhouse climate control—to illustrate its practical value.

- Response to Comment 5 (Revision): We have defined digital twin and provided more examples.
- Revisions made correspond to lines 725~728 in the manuscript.
- The follow references were added, to provide more concrete explanation/examples:
 - [155] A single view leaf reconstruction method based on the fusion of ResNet and differentiable render in plant growth digital twin system. Cited in line 729
 - [156] A 3D functional plant modelling framework for agricultural digital twins. Cited in line 735

Comment 6: Please replace every instance of "Lindemyer-systems" with "Lindenmayer systems (L-systems)" and cite the original 1968 paper.

- Response to Comment 6 (Revision): Good catch, thank you, we have made the changes.
- Revisions made correspond to lines 507, 509 in the manuscript.
- The reference to the original 1968 paper was added [116] Mathematical models for cellular interactions in development I. Filaments with one-sided inputs. Cited in line 513

Comment 7: A few statements about the importance of spatiotemporal modeling appear both in the Abstract and again in the Introduction. Trimming those duplicates will tighten the writing and avoid repetition.

- Response to Comment 7 (Revision): We have gone through and trimmed some redundant statements throughout the paper.

Comment 8: In the "Scope of this Review" section or in the Conclusion, you might add a clarifying sentence such as: "This review focuses on controlled-environment agriculture from 2015-2025; open-field models and pre-2015 foundational work are beyond our scope." That will set clear boundaries for readers.

- Response to Comment 8 (Revision): We have a similar statement in the literature overview section 2.1. We have expanded the statement to make it more verbose. Revisions made correspond to lines 182~185

Comment 9: All existing figures and tables are valuable, so please retain them. To implement the comparative synthesis suggested above, add a new table rather than removing any current illustrations.

- Response to Comment 9 (Revision): We have retained the figures, with slight modifications made based on comments from the other reviewers. I have added new tables (Table 3 & 4) based on your suggestions
- **Revisions made correspond to lines in the manuscript:**
 - Section 3.3: lines 448~461
 - + Table 3: Page 17
 - Section 4.3: lines 624~639
 - + Table 4: Page 25

To Reviewer #3:

Comment 1: In the title and many places in text, the paper highlights the term "representations". However, the review is mostly about models and approaches. Please clarify what "representations" mean or revise this term (recommended).

- Response to Comment 1 (Revision): We have added a subsection 1.2 Defining Simulated Environment and Plant Growth, where we clarify some meaning of terms throughout paper including representations.
- Revisions made correspond to lines 66~91 in the manuscript.

Comment 2: I would suggest to revise the title to "Spatiotemporal Modeling of Plant Growth Prediction in Simulated and Controlled Environments: A Comprehensive Review"

- Response to Comment 2 (Revision): We agree the title may be too confusing and may not fully represent them. However we do spend some time discussing various representations in 2D/3D and as functional models. We will compromise and change the title to: Predictive Modeling, Pattern Recognition, and Spatiotemporal Representations of Plant Growth in Simulated and Controlled Environments: A Comprehensive Review
- Revisions made correspond to lines 1~3 in the manuscript.

Comment 3: I suggest remove the datasets parts and Table 2. It distracts the flow of the reading. The paper is mainly about the modeling.

- Response to Comment 3 (Rebuttal): Comments from reviewer 2 have suggested we do not remove any figures. We also believe this table adds value to discussions to section 5.2 Dataset Gap and to overall discussions on model representations. We will leave the table for the time being, but understand your concerns.

Comment 4: Section 2.2 is better placed ahead in Introduction.

- Response to Comment 4 (Revision): We have moved this section and it is now section 1.4.
- Revisions made correspond to lines 143~159 in the manuscript.

Comment 5: In Figure 1, the use of lowercase and uppercase letters should be kept consistent.

- Response to Comment 5 (Revision): We have fixed capitalization of the figure to make it consistent. We only left minor words and hyphenated words lowercase.
- Revisions made correspond to Figure 1 on page 4.

Comment 6: In Figure 2, it would be good to list the publication year and venue of the models and datasets. The datasets and approaches are better shown in different plots.

- Response to Comment 6 (Revision): We have separated the methodologies and datasets into two different plot.
- Revisions made correspond to Figure 2 on page 7.

Comment 7: The categorization of regression models (3.1) between latent representation models (3.2) is not entirely convincing, because a latent representation model can also be regression-based. Please clarify.

- Response to Comment 7 (Revision): We understand your point. In this case, we believe a better term would be indirect latent representation, or hidden state representation, since scalar quantitative traits have a functional meaning while the former is some abstracted representation with no direct functional meaning. The latent representations are a lower dimension form of quantitative traits through time, so the dimensionality reduction considers a compounding of QT with time dimension. We have added a footnote clarifying the meaning of latent representation in this context, in the introduction of section 3.2
- Revisions made correspond to line 323 and footnote 1 on page 12 of the manuscript.

Comment 8: Can a potential formulation for plant growth prediction be provided in Transformer and attention-based modeling.

- Response to Comment 7 (Rebuttal): Since we could not find high quality works that utilize transformers and attention networks within the context of spatiotemporal analysis of plant growth, I have opted not to get into depth in this topic but to mention as a potential next step within this domain.

Comment 8: L108, classification tasks -> recognition tasks

- Response to Comment 8 (Revision): Correction has been made.
- Revisions made correspond to line 174 of the manuscript.

Comment 9: L468, recent work have -> recent work has

- Response to Comment 9 (Revision): Correction has been made.
- Revisions made correspond to line 522 of the manuscript.